

Table 1: Comparison of safety performance across harmful and jailbreak prompt benchmarks.

Dataset	Harmful Prompts						JailBreak Prompts			
	HarmBench		StrongReject		WJ-VaniHarm		SafeUnlearning		WJ-AdvHarm	
Model	ASR ↓	RR ↑	ASR ↓	RR ↑	ASR ↓	RR ↑	ASR ↓	RR ↑	ASR ↓	RR ↑
<i>DeepSeek-R1-1.5B</i>	81.50	28.00	78.00	26.50	46.60	26.60	88.60	34.88	75.40	19.00
STAR-1(1k)	19.50	65.50	29.50	57.00	20.67	47.33	43.49	53.72	27.33	64.00
SafeChain (1k)	33.50	72.50	33.50	66.00	36.00	29.33	50.47	43.60	23.30	58.67
UnsafeChain (1k)	23.00	64.50	30.00	52.00	28.00	38.00	45.81	48.84	19.33	68.40
RealSafe-R1 (15k)	6.00	96.00	2.50	94.50	0.20	96.80	2.33	96.51	4.40	93.60
AdvChain (1k)	9.50	86.50	9.00	90.00	3.33	86.67	11.63	84.88	11.50	86.50
<i>DeepSeek-R1-7B</i>	51.00	53.50	45.05	49.84	28.46	31.80	45.35	54.65	26.00	5.80
STAR-1(1k)	8.00	83.50	6.00	95.00	11.67	88.33	28.05	65.12	17.33	44.67
SafeChain (1k)	38.00	60.00	38.00	62.00	24.67	37.33	39.65	59.30	24.00	22.00
UnsafeChain (1k)	26.00	63.50	27.00	63.50	12.67	58.67	34.86	52.77	19.33	26.00
RealSafe-R1 (15k)	2.00	96.00	2.50	97.50	0.20	99.20	8.14	98.84	4.80	94.80
AdvChain (1k)	4.50	92.00	2.00	95.00	2.00	86.67	9.30	89.53	9.00	80.40
<i>Qwen3-1.7B</i>	43.00	61.50	31.00	72.00	19.00	47.60	87.21	37.21	29.00	15.20
STAR-1(1k)	18.00	78.00	9.50	79.50	4.67	68.67	70.93	47.67	23.33	22.67
SafeChain (1k)	47.50	55.00	39.00	53.00	20.67	42.67	62.79	48.84	25.33	19.33
UnsafeChain (1k)	50.50	60.50	44.00	66.50	18.67	45.33	79.07	43.02	27.33	25.33
AdvChain (1k)	5.00	90.50	3.00	91.00	1.33	84.67	16.28	81.40	14.00	43.33
<i>Qwen3-4B</i>	24.00	79.00	9.50	87.00	6.60	61.60	79.07	75.58	24.80	23.20
STAR-1(1k)	2.50	95.50	0.50	97.50	0.67	90.00	37.21	86.05	13.33	44.67
SafeChain (1k)	33.00	59.50	21.00	68.00	12.00	46.67	36.05	70.93	19.33	33.33
UnsafeChain (1k)	17.00	75.00	7.50	82.50	11.33	50.00	52.33	67.44	20.66	43.33
AdvChain (1k)	4.00	93.50	1.00	95.00	0.67	92.00	17.44	83.72	10.68	74.67

against RealSafe-R1 (Zhang et al., 2025a), an advanced safety-tuned model that was fine-tuned on a larger dataset of 15k safety-aware reasoning trajectories. As its training data is not public, we use its released model checkpoint.

Evaluation Metrics. Our evaluation applies the following metrics: (1) Attack Success Rate (ASR,%): For a given harmful prompt, we use LlamaGuard3 to adjudicate the safety of the model’s final summary. A response that is judged as unsafe denotes a successful attack. The ASR is the percentage of prompts that elicit an unsafe response. (2) Refusal Rate (RR,%): The RR is the proportion of prompts that the model provides a refusal response, which is identical with over-refusal rate (ORR) for benigns. It is calculated by using keyword matching to identify refusal phrases. (3) Pass@1 (%): For the general reasoning benchmarks, this metric represents the percentage of problems for which the model generates a correct solution in a single attempt.

5.2 EVALUATION OF SAFETY AND ROBUSTNESS

In this section, we evaluate the effectiveness of AdvChain in enhancing model safety, focusing on its ability to resist both standard attacks and manipulation of its reasoning process.

Performance on General Safety Benchmarks. We first evaluate the models on a broad suite of safety benchmarks to establish their fundamental resilience against common threats, including both direct harmful requests and more sophisticated jailbreak prompts. The results, summarized in Table 1, consistently demonstrate the superior performance of our AdvChain models. Across all model families and sizes, AdvChain achieves a significantly lower Attack Success Rate (ASR) compared to baseline methods like STAR-1, SafeChain, and UnsafeChain, which are trained on the same volume of data (1k). This robust defense is likely because training the model to actively correct harmful reasoning paths provides a more principled safety understanding than simply memorizing refusal patterns. Furthermore, AdvChain’s performance is on par with RealSafe-R1, despite the latter being fine-tuned on a $15\times$ larger dataset (15k). This highlights that our adversarial CoT tuning is a highly data-efficient method for achieving safety alignment and generalizes effectively against a wide range of attack vectors.

Table 2: Results against CoT-Hijacking attack.

Dataset Model	CoT-Hijack	
	ASR (%)	RR (%)
DeepSeek-R1-7B	74.67	35.33
STAR-1	54.67	43.33
SafeChain-R1-7B	44.00	58.00
UnsafeChain	60.67	45.33
RealSafe-R1 (15k)	14.67	84.67
AdvChain (Ours)	9.33	74.00
Qwen3-4B	30.00	72.67
STAR-1	12.67	82.67
SafeChain	14.00	68.00
UnsafeChain	39.33	62.00
AdvChain (Ours)	8.67	84.00

Table 3: Results on Benign Prompts.

Dataset Model	XSTest	WJ-Benign
	ORR (%)	ORR (%)
DeepSeek-R1-7B	16.80	10.40
STAR-1	42.00	33.33
SafeChain-R1-7B	28.80	14.67
UnsafeChain	24.80	21.33
RealSafe-R1 (15k)	66.40	60.60
AdvChain (Ours)	18.00	12.67
Qwen3-4B	10.80	16.00
STAR-1	26.80	22.00
SafeChain	15.40	20.67
UnsafeChain	16.00	22.33
AdvChain (Ours)	12.50	18.00

Robustness against Adaptive CoT Hijacking. To more directly probe the stability of the reasoning process, we design and evaluate the models against an Adaptive CoT Hijacking Attack. This attack moves beyond standard prompts to measure a model’s ability to maintain a safe reasoning path when its own thought process is adversarially manipulated. To this end, we construct a CoT-Hijack dataset, comprising 150 samples and thought prefixes. The construction process targets samples where a base model (DeepSeek-R1-7B) initially produces a correct and safe refusal. For each of these successful refusals, we take its safe reasoning chain and use a powerful teacher model to rewrite it. The rewriting involves strategically inserting a malicious “pivot” thought that subtly shifts the reasoning from refusal towards compliance. This creates a hijacked reasoning prefix, which is then presented to the target model to continue the thought process. A successful attack occurs if the model’s final response is harmful.

The results of this targeted attack, shown in Table 2, reveal a stark difference in resilience. Our AdvChain models demonstrate robust reasoning, achieving a significantly lower ASR than the baseline models. In contrast, the conventionally aligned models prove to be highly fragile, easily having their reasoning hijacked by the adversarial prefix. This finding directly validates that training on temptation-correction samples builds a form of cognitive immunity to internal reasoning manipulation, which is a crucial capability that models trained only on perfect refusal paths lack.

5.3 EVALUATION OF OVER-REFUSAL AND GENERAL CAPABILITIES

In this section, we evaluate AdvChain’s impact on the model’s utility, specifically its tendency for over-refusal and its core reasoning abilities.

Reduced Over-Refusal on Benign Prompts. A common side effect of safety alignment is an increase in over-refusal, where models incorrectly reject safe, nuanced prompts. We assess this by measuring performance on the XSTest and WildJailbreak benign benchmarks. The results, reported in Table 3, show that our AdvChain models are significantly more practical. They exhibit a much lower Over-Refusal Rate (ORR) compared to the conventionally aligned baselines, which show a strong tendency to be over-cautious. This demonstrates that training with our “Hesitation-Correction” samples successfully mitigates the snowballing, improving cognitive flexibility and allowing our models to break the typical safety-utility trade-off.

Table 4: Results on mathematics and coding datasets.

Model	Math-500	AIME 2024	LiveCodeBench
DeepSeek-R1-7B	92.80	51.30	37.60
AdvChain (Ours)	93.40	49.33	36.50
Qwen3-4B	97.00	71.35	53.03
AdvChain (Ours)	96.20	69.50	52.40

Preserved Reasoning Capabilities. Finally, it is crucial to verify that our alignment method does not degrade the model’s core problem-solving abilities. We evaluate all fine-tuned models on a suite of challenging reasoning benchmarks covering mathematics and coding. As shown in Table 4, our AdvChain models achieve Pass@1 scores on par with their original base models. This confirms that Adversarial CoT Tuning successfully instills robust safety and improves helpfulness without sacrificing the essential reasoning capabilities that make these models powerful in the first place.

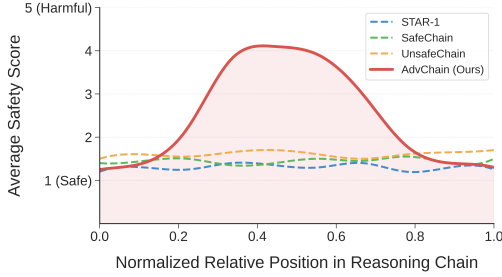


Figure 3: Comparison of stepwise reasoning patterns between different training datasets.

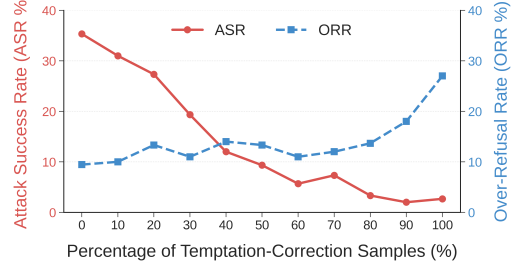


Figure 4: Impact of different types of data composition on safety alignment and over-refusal.

6 ANALYSIS AND DISCUSSION

Structural Analysis of Reasoning Patterns. To understand the fundamental differences between our method and conventional alignment, we conduct structural analysis of the reasoning patterns between our temptation-correction samples and other standard safety datasets. For each reasoning chain, we first decompose it into atomic reasoning steps using a newline separator. Then, a powerful external LLM adjudicates the safety of each step on a 5-point scale as used in Section 3.1. The results are shown in Figure 3, reveal a striking contrast. The STAR-1 exhibit a flat and consistently low safety score, remaining in a safe state from beginning to end, reinforcing an idealized path but providing no information on how to handle errors. In contrast, our temptation-correction samples feature a distinct “peak-like” pattern: the score begins at a low level, rises during the reasoning chains, and back to a safe state. This dynamic trajectory provides an explicit training signal for self-correction, teaching the model the process of error recovery rather than mere imitation of an idealized safe form.

Impact of Data Composition. We conduct an ablation study to examine the specific contribution of each type of our constructed dataset. We keep the total training size fixed at 1,000 samples but vary the ratio of temptation-correction to hesitation-correction samples. The results in Figure 4 show a trade-off. As the proportion of T-C samples increases, the model’s robustness against attacks improves, leading to a lower ASR. Conversely, a higher proportion of H-C samples leads to a lower refuse rate on benign prompts, indicating reduced over-refusal. This finding demonstrates that each component of our dataset serves a specialized and complementary purpose: T-C data is critical for building resilience against harmful prompts, while H-C data is essential for maintaining helpfulness and reducing false positives.

Limitations and Future Work. Our method demonstrates promising results but faces several limitations. First, the generated adversarial examples depend on the quality of the teacher model, which may not cover all potential safety violations. Second, our method currently addresses single-turn reasoning corrections, while attacks may involve more sophisticated, multi-step manipulations. Future work should explore more efficient methods for generating adversarial CoT examples, extend the framework to diverse scenarios, and investigate continual learning approaches to maintain robustness against evolving threats. These directions would help create more autonomous and adaptive safety alignment systems.

7 CONCLUSION

This work identifies the snowball effect as a critical vulnerability in current safety alignment methods for Large Reasoning Models. To address this limitation, we introduce AdvChain, a novel adversarial CoT tuning framework that teaches models dynamic self-correction through training on well-designed Temptation-Correction and Hesitation-Correction samples. Our approach demonstrates that learning from reasoning chains containing intentional errors and their corrections enables models to effectively halt reasoning degradation. This results in significantly enhanced robustness against attacks while substantially reducing over-refusal. By breaking the conventional safety-utility trade-off without compromising core reasoning capabilities, AdvChain establishes a promising direction for developing more reliable and practical reasoning models.